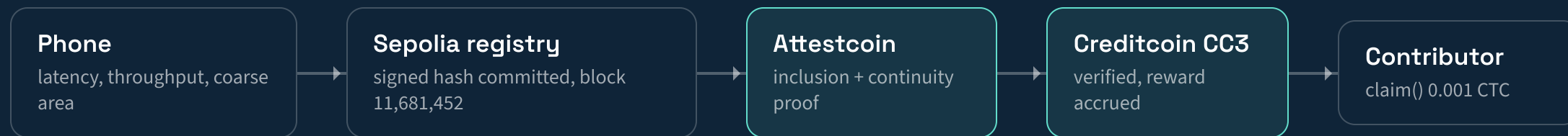




SIGNALPROOF

Connectivity data a contract can check.

Phones measure real network conditions. Each measurement is committed on Ethereum Sepolia, proven by the Attestcoin Protocol, and settled with a reward on Creditcoin.



BUIDL CTC 2026 Fall DePIN track Deployed and verified on Sepolia and CC3 Testnet



Anyone can claim coverage numbers. Nobody can prove them.

Operators publish coverage maps. DePIN networks pay contributors for connectivity data. Venues and public programs buy both. None of them can audit what they are paying for.

The data is self-reported. A coverage claim is a number someone typed. The device that produced it, the place, the time — none of it is checkable after the fact.

The incentive is to inflate. Where contributors are paid per sample, a fake sample is worth exactly as much as a real one until someone can tell them apart.

Location data is the price of honesty today. The usual fix is to collect precise GPS trails, which no one should have to hand over to prove their phone had two bars.

What SignalProof changes

The measurement becomes an event on a public chain, the event is proven to a second chain by the Attestcoin Protocol, and a contract there decides — from the proof alone — whether it is real enough to pay for.

Not the gateway. Not the relay. Not the person who ran the test. The proof.

One measurement, two chains, one decision.

Everything before the Sepolia event decides what is **admitted**. Everything after it decides what is **paid**. The two never trust each other.

Measure

Browser client: median of 7 timed pings, a 3 MB download, the network class, and a geohash-6 area. No coordinate leaves the device.

seconds

Admit

Gateway recomputes the measurement root, recovers the contributor's signature, checks freshness and uniqueness, then the relayer commits the hash to SourceBatchRegistry.

next 15 s tick

Attest

Creditcoin's attestation of Sepolia catches up to the block. The worker polls the attested height once per tick instead of blocking.

7–9 min

Prove

ProofBuilder returns the Merkle inclusion proof and the continuity proof for the transaction.

7 siblings, 6 roots

Settle

SignalProofSettlement verifies through the BlockProver precompile, applies its own checks, and accrues 0.001 CTC to the contributor.

150,640 gas

Recorded end to end: 8.5 minutes, from the Sepolia transaction to a claimable reward. The attestation wait is the whole cost; everything else is seconds.

Payment is permissionless. Anyone holding a valid proof may settle it. All authorisation lives in the contract's checks, so a relayer that disappears cannot strand a measurement.

What the precompile proves, and the two things it does not.

SignalProofSettlement inherits ASCBase and overrides one hook. The BlockProver precompile proves inclusion in an attested Sepolia block — not that the transaction succeeded, and not which contract emitted its logs.

Protocol detail	What we do
Receipt status	Reject unless <code>receiptStatus == 1</code> . Inclusion is not success.
Emitter binding	Reject a log whose emitter is not our registry. Missing from the docs' SimpleMinterASC example, present in Gluwa's ASCLoanManager; mirrored here with a whole-system test where a forged registry holds a valid proof.
Contributor signature	Recovered on-chain by the registry from the exact EIP-191 text the wallet signed. A relayer that names another payee holds no signature for that pairing.
chainKey	Resolved at startup from <code>getSupportedChains()</code> on <code>chainId</code> . It is a <code>uint64</code> ; <code>chainName</code> is hex-encoded; key 1 means another chain on Mainnet.
Attestation wait	Poll the attested height once per tick; the SDK's blocking wait would freeze the queue for 8–20 minutes.

Two replay layers

ASCBase deduplicates by `queryId`; the contract also refuses a `measurementRoot` it has already settled.

Batch route

Many measurements, one continuity proof, one transaction. A bad entry is skipped and reported, never fatal to the honest ones beside it.

Pull-payment rewards

No value moves on the verification path, so a reverting receiver cannot block settlement for anyone else.

Deployed, verified, and settled on-chain.

Contract	Chain	Address
SourceBatchRegistry (signature on-chain)	Sepolia 11155111	0x32c0923cD58523864D2727FCaaB109783664c236
SignalProofSettlement	CC3 Testnet 102031	0x8F14B2cC1b807203d332DE6E3DA6274176FDb584
SignalProofBatchSettlement	CC3 Testnet 102031	0x3B90e22f246bBa68f6de682b564c33b121D68C85

First run through the signed registry	Result
Signed measurement committed to Sepolia	0xbd7261...7223 block 11,681,452
Attestation reached that block	9.6 min, advancing 10 blocks at a time
Inclusion proof	7 Merkle siblings, 9 continuity roots, 2,112-byte txBytes
Settled on Creditcoin	0x60d9e4...a2b5 block 5,468,910, 154,224 gas
Earlier: claim()	0x148536...e0030 pool 5.000 to 4.999 CTC

9.8

minutes end to end

8.5 on the earlier run; the attestation wait is the whole cost

3 in 1

batch settlement

Three measurements, one continuity proof, one CC3 transaction — 0xcf1a01...c49598 and 0xb0933e...32d990.

75

Foundry tests

127

server tests

plus a live smoke test of the Attestcoin read path that passes from a clean clone.

Evidence — all transactions are linked in the README

SignalProof

Four ways to get paid for nothing, found and closed before submission.

Each exploit is kept in the test suite with its assertions inverted, so reopening the hole fails a test instead of quietly working.

1

The registry was permissionless

Anyone could call the real registry, name themselves as payee, obtain a genuine proof and settle it. Emitter binding passes, because the event really came from the bound registry. Fixed by gating admission to the relayer that carries the gateway's checks. Redeployed.

`AdvUnlimitedMint.t.sol`

2

One measurement paid twice

Two settlement routes, one registry, two `settled` maps. The same genuine proof could be paid from both pools. The batch contract now defers to any root a sibling already paid, via a guarded `staticcall` that degrades to no cross-check rather than refusing everyone.

`DoubleSettlement.t.sol`

3

One poisoned log killed a batch

A Sepolia transaction that touched the registry and emitted a lookalike log made the batch revert forever, taking the honest measurements down with it. Foreign logs are now skipped and reported through `ForeignLogSkipped`. `BatchPoison.t.sol`

4

The relayer could name the payee

Gating on the relayer left one party fully trusted: it could commit a genuine event naming anyone as contributor. The registry now recovers the contributor's EIP-191 signature on-chain; a vector signed with ethers is accepted by the contract.

`SourceBatchRegistry.t.sol`

Why this matters for Attestcoin depth. The precompile was never wrong in any of these; every exploit used a genuine proof. Admission, attribution, replay and batching are where cross-chain security is won or lost — that is where this project spent its time.

Batching is cheaper per entry, and it gets more expensive as it grows.

Two live runs are not enough to state a saving, so `BatchGasProbe.t.sol` holds the proof shape fixed and varies only the batch size.

Entries	Total gas	Marginal gas per extra entry
1	77,828	—
10	482,141	44,892
50	2,468,789	51,015
200	12,476,653	71,165

A marginal entry costs **45–71k gas** against 78k for a batch of one — and the number rises with size, because the decoder allocates memory it never reclaims and the quadratic memory-expansion term takes over.

So the worker caps a batch at 50, not at the block ceiling. The probe puts that ceiling at 600 entries for a 75,000,000-gas CC3 block; 800 does not fit.

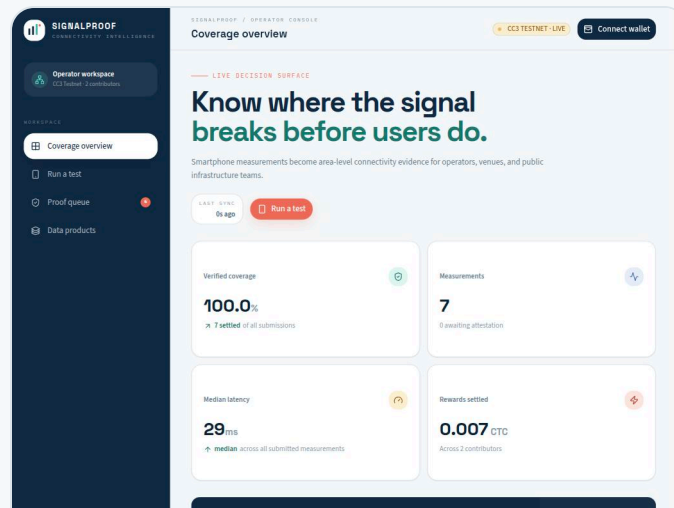
Live numbers agree in shape: 253,284 and 423,360 gas for three measurements, against 150,640 for one. The spread comes from the proof the attestation service happens to return, not from the batch.

Measured, not estimated

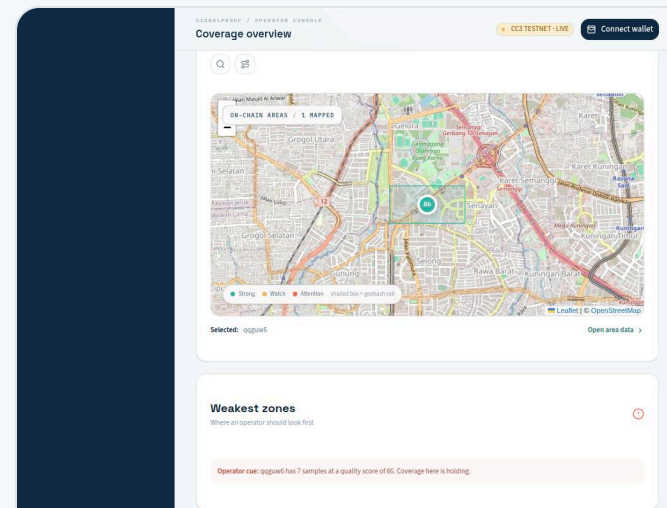
SignalProof

The dashboard reads the chains, not a database.

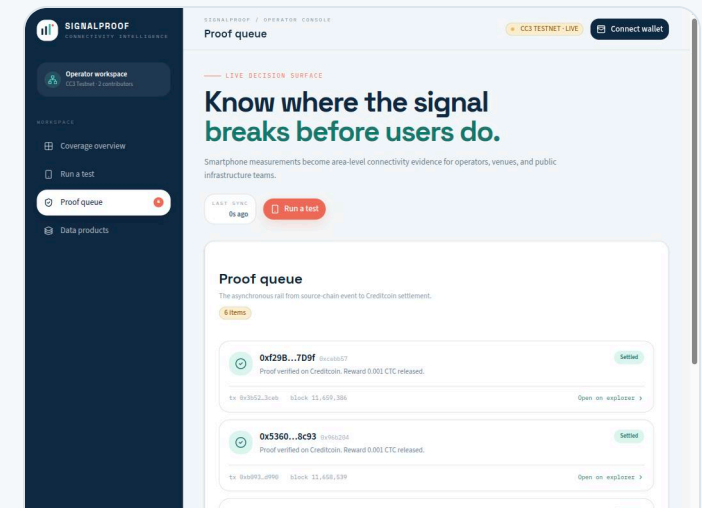
Every number below is a join of `MeasurementSubmitted` on Sepolia with `MeasurementVerified` on Creditcoin. A keyless clone shows the same thing, and a buyer can query it: `/v1/areas/qqguw6` returns the aggregates and every sample with its two transactions.



Coverage overview. Verified share, settlements, rewards, and the four-stage proof spine, all live.



Coverage map. The shaded box is the geohash cell itself, 1.2 km by 0.6 km. Undecodable areas are listed as unmapped, never placed.



Proof queue. Each row opens the live Attestcoin proof — attested height, Merkle path, continuity roots, the exact `execute()` call — and can be settled from the contributor's own wallet.

Coarse by construction. Measured, not asked.

Location never leaves the phone

The area is a geohash of at most six characters, enforced by the gateway, stored on-chain as `bytes32`. A cell is roughly **1.2 km by 0.6 km**; a contributor is somewhere inside the box, never at a dot. The raw coordinate is converted in the same function that reads it and is never transmitted.

The wallet is an identity, not a signer

The relayer pays gas on both chains. The connected address only decides who accrues the reward. The one transaction a contributor ever sends is `claim()`.

Why we measure instead of asking the browser

	Measured by us	Reported by <code>navigator.connection</code>
Throughput	40.8 Mbps	7.5 — capped at 10 by Chromium
Latency	68 ms median, 7 samples	50 — rounded to 50 ms
Carrier	not collected	no API exposes it
5G	not claimed	no such value exists

The UI used to show “5G · Telkomsel”. Both halves were fabricated, so both were deleted. What no browser can provide, the product does not display.

What is not done, stated before a judge finds it.

Known limitations

- The relay is still the admission point: if it goes offline, new measurements wait. It can no longer forge attribution — the registry checks the contributor's signature on-chain.
- Read direction only, Sepolia to Creditcoin. Write-ability has no public reference implementation and has not cleared third-party audit.
- Attested is not finalized: attestation runs 65–85 blocks ahead of Sepolia's `finalized` tag. A deep reorg could in principle invalidate an attested height.
- End to end is 9–13 minutes. A live real-time demo is not possible; the demo pre-warms a proof.
- Anti-Sybil is a rate limit (3 per contributor per cell per 10 min), not a defence; no platform attestation; the browser client is the measurement client. The buyer API has no authentication or retention policy.

Next 90 days

- Buyer API v2: authentication, retention policy, and cell-level subscriptions on top of the read-only v1 that ships today.
- Native measurement client with background sampling and device attestation.
- Sybil resistance beyond the rate limit: device attestation, stake-weighted rewards, cell-level anomaly detection.
- Write-ability once an audited reference implementation exists, so settlement can push results back to the source chain.

Why this belongs on Creditcoin.

The buyer is real. Operators planning capacity, venues negotiating with carriers, and public infrastructure programs auditing coverage commitments all pay for connectivity data today, and none of it is verifiable.

The proof is the product. A coverage snapshot whose every sample can be traced to an attested Sepolia event and a Creditcoin settlement is a different good from a spreadsheet of claims.

Creditcoin is where it settles. Rewards accrue in CTC, the settlement contract is the source of truth, and the Attestcoin Protocol is the reason no oracle operator sits in the middle.

Links

Repository	github.com/mdlog/signalproof
Demo video	linked from the BUIDL page on DoraHacks
Settlement contract	creditcoin-testnet.blockscout.com/address/0x8F14...b584
Source registry	sepolia.etherscan.io/address/0x32c0...c236
Protocol	docs.attestcoin.org

Built on the Attestcoin Protocol by Gluwa. Contract patterns follow [gluwa/attestcoin-protocol-examples](#) (Apache-2.0). MIT licensed.